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Theory of Mind and Conceptual Change

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We agree with the commentaries by Scholl and Leslie, and also by Moses, that the meta-analytic findings do not definitively rule out early competence accounts. But they do make extant versions of such accounts increasingly unlikely. In particular, the meta-analytic findings argue against executive function expression accounts, including the Theory-of-Mind Mechanism/Selection Processor account advocated by Scholl and Leslie. Specifically, Scholl and Leslie articulate two explicit predictions of their account: that task manipulations that attenuate inhibitory demands should differentially advantage older children, and that theory-of-mind developments should occur with consistent timetables. Both of these specific predictions are clearly contradicted, not supported, by the meta-analytic findings.

INTRODUCTION

Getting the data right and getting the theories right, are two important and intertwined tasks for any scientific endeavor. On this we and the commentators agree. It is especially pleasing, therefore, that the commentaries mostly concur that our analyses (Wellman, Cross, & Watson, 2001) have helped to "get the data right" regarding an important developmental acquisition—children's understanding of false beliefs.

COMBINED MODELS

Moses (2001, p. 689), however, notes that in our combined model, the fact that the best-effects line fails to show a different slope from the other lines "would appear to be a mathematical necessity rather than an empirical finding." His assertion is both true and not true, and we welcome this opportunity to clarify.

In the testing of combined models we examined several different models. As reported in our article (Wellman et al., 2001), we examined models with a variety of factors, but then pruned those factors to the nonredundant set that best predict the data. We then used that final model—which included only the factors of deception, real presence, and salience, along with age—to generate the best-effects trajectory (as well as no-effects and worst-effects trajectories) graphically depicted in Figure 10 of our article. Thus, we engaged in a two-step process: (1) exploring and evaluating alternative models to arrive at a final model, and (2) using the final model to predict the effects of combining the key enhancing factors.

In our article we did not completely report the variety of models that were tested. However, we considered combined models that included interaction terms as well as models that included only simple additive factors. Moses is correct that our attempts to do this were constrained by the data themselves—some of the critical cells were indeed empty (because, as Moses, p. 689 notes, "no studies manipulate all the enhancing factors"). Thus, we never tested a model with a three-way interaction term (e.g., Salience × Real Presence × Deception) because of the problem Moses indicates, namely the occurrence of empty cells. We did, however, test models that included many two-way interaction terms.

Consider the following critical comparison: (1) a model that includes deception, real presence, and salience, along with age as additive factors (the final model reported in our article) versus (2) a model that includes those factors as well as all possible two-way interactions; namely, Salience × Real Presence; Real Presence × Deception; and Salience × Deception. Empirically, it turns out that the first model, as already reported, produced an R^2 of .460 (accounting for 46% of the variance); the second model produced an R^2 of .478. The difference in amount of variance accounted for, in comparing these two models, was thus .018, $F(7, 348) = 1.67$, *ns*. That is, the second model provided no improvement in fit over a model with no interaction terms. At the same time, the second model was a justifiable model because it involved no cells with zero conditions (indeed, the smallest cell size was 5 in the marginal tables for that model).

Another model that we tested was one that included three interaction terms that represented all the interactions between age and the three categorical predictors of salience, real presence, and deception. That model yielded an R^2 of .470; comparing it to the first model yields, $F(5, 350) = 1.3$, *ns*.

Response to commentaries on Wellman, Cross, & Watson, "Meta-Analysis of Theory-of-Mind Development: The Truth about False Belief."

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In sum, a model that includes these higher order interactions is unnecessary to account for the available data; that is, the model we termed our final model in the article—which included only additive effects—was completely sufficient to model the data. Thus, the best-effects line depicted in Figure 10, was in fact mathematically constrained just as Moses asserts. But, by the time the best-effects prediction was generated, constraining that prediction in this fashion was appropriate because the underlying model—an additive one without additional interactive terms—best accounted for the data.

What about Moses's (2001, p. 689) claim that our "estimates of such effects are extrapolations that go well beyond the data"? To be clear, the estimates of the additive effects that yielded the best-effects predictions were not themselves extrapolations of the data, but were estimated from subclassifications of the data. Further, the lines plotted were not extrapolations either, because they were only generated for the ranges of age and conditions included in the meta-analytic data base. However, it is true that the best-effects combination was a prediction, and in that sense an extrapolation, because no extant study has actually tested a combination of these three enhancing manipulations. The predicted line shown in Figure 10 of our article thus did go "beyond the data" in part, but this is true of all good predictive modeling. More important, we believe, is the fact that the combined model featured as our final model was conservative and robust and so provides a statistically well-grounded platform for such predictive modeling. In particular, it had the advantage of utilizing the information available in 362 different conditions—a comprehensive compilation of a large majority of the relevant, available data generated by numerous investigators across numerous investigations.

To reiterate the conclusion we arrived at in our target article, then, the meta-analysis demonstrated a consistent age-related transition from below-chance to above-chance performance roughly appearing between 3 to 5 or 6 years of age. This transition appeared across numerous task variations including myriad task simplifications designed to enhance the performance of young children by minimizing a variety of task demands. Although Scholl and Leslie (2001, p. 700) claim that this finding is nothing more than an "expected robust age effect in all of the meta-analytic results," we believe this is an important empirical contribution, as do three of the four commentaries. In advance of our meta-analysis, it was certainly conceivable that the data would have yielded a different pattern of results. Indeed, several authors have plausibly claimed that the data from their individual studies should best be interpreted as consistent with a larger pattern of results that would be like those in Figure 3C (Wellman et al., 2001). In contrast, the meta-analysis showed that the data are demonstrably like those in Figure 3B.

Our hope was that clarifying the empirical data would set the stage for more informed theorizing. That hope seems to be already partially fulfilled because most of these commentaries are full of thoughtful discussions of the significance of our findings along with discussions that expand the implication of these findings in intriguing new directions. We will say little more about these creative extensions, other than to voice appreciation for such insightful theorizing. Of course, setting the stage for informed theorizing also means setting the stage for theoretical debate of the type provided by Scholl and Leslie. Therefore, the remainder of our response will focus primarily on addressing the theoretical challenges posed by that commentary.

THEORY OF MIND MODULE

Scholl and Leslie (2001) identify several instances in which they believe we have misunderstood their Theory-of-Mind Mechanism/Selection Processor (ToMM/SP) theory.[1] Complementarily, they seem to have misread our claims as well. Most generally, they seem to assume that when we concluded that the meta-analysis supports conceptual change positions, that we identify conceptual change in general with the type of conceptual change specified in a theory-theory (Gopnik & Wellman, 1994; Wellman & Gelman, 1998). We do not.

At the most general level we argue only that the data provided evidence of some sort of conceptual change in children's understanding. At this general level of analysis, modular mechanisms could certainly lie behind children's changing conceptions. For example, in Leslie's larger theorizing he posits a series of (at least two) modules (see also Baron-Cohen,

[1] At the same time, Scholl and Leslie either clearly misunderstand theory-theory accounts (e.g., Gopnik & Wellman, 1994) or only wish to deal with an easy-to-reject strawman version of that position. In our theory-theory accounts, for example, none of the following are true, although Scholl and Leslie attribute such claims to us:

1. Four-year-olds have reached the end of the line developmentally (to the contrary, see Wellman, 1990; Wellman, in press; Wellman & Hickling, 1994).
2. The processes that account for conceptual change are completely general (to the contrary, see Wellman & Gelman, 1992, 1998).
3. A concept of belief is learned "from scratch" (to the contrary, see Bartsch & Wellman, 1995; Gopnik & Wellman, 1994).

1995). He argues that, at first, children reason about human behavior via a theory of body module (ToBY), and only later does ToMM come on line, thereby providing a construal of human action in terms of mental states (Leslie, 1994). In this transition from ToBY to ToMM, there is conceptual change in the larger sense; a new type of construal of human action now appears. The cause of such conceptual change, in this proposal, is the activation of an additional innately based module. If Leslie's theory posited that this change takes place at about age 4 years or so, then the meta-analysis would have provided evidence for that type of conceptual change (just as it provided support for other claims about conceptual change at this age, such as those in theory-theory). However, no extant position posits a modular change of this type, that is, a new module coming on line at 4 years. In particular, Leslie's (1987, 1994) position is that ToMM comes on line at 1½ or 2 years of age, and that changes from 2 to 6 years instead involve changes in SP, a component that is supplemental to ToMM (or ToBy) and that (according to Scholl and Leslie, 2001, p. 697) is "conceived to be a general executive process required in many situations." So we turn now to the specific challenges Scholl and Leslie offer to our interpretation of the meta-analytic findings, challenges based on Leslie's specific ToMM/SP account (Leslie, 2000; Leslie & Polizzi, 1998).

Scholl and Leslie claim, contrary to our interpretation, that the meta-analytic pattern of findings does not favor a conceptual change interpretation over an early competence one. Specifically, they argue that Leslie's ToMM/SP model—a modular early competence position—is also consistent with our findings.

A primary claim that they make is that our findings do not definitively rule out early competence interpretations. For example, Scholl and Leslie (2001, p. 698) note that the "particular task manipulations considered in the meta-analysis did not sufficiently attenuate" young children's performance problems with false-belief tasks. If not, those manipulations would not sufficiently enable young children's competence to reveal itself in consistently correct judgments. As they argue (2001, p. 698), "there may even be no combination of manipulations that makes the task easy enough for 3-year-olds." This is a general point that is also made by Moses (2001) and with which we wholeheartedly agree: If young children fail on a task, no matter how carefully constructed, this does not definitively demonstrate that they lack the critical underlying understanding. It is always possible that they possess the underlying understanding, but the task has just simply failed to successfully reveal it. Logically, no amount of empirical data, no matter how voluminous, can definitively rule out the possibility of an as-yet-unrevealed underlying competence.

What an increasing amount of data, exploring a variety of creative task variations, can do, however, is make such a possibility increasingly unlikely. The meta-analysis encompassed such a large variety of variations, simplifications, and controls that it substantially increased the likelihood that young children's difficulties reflect difficulties with the conceptual heart of the task, beyond any difficulties with particular task requirements. While not definitive, the meta-analysis provides this conclusion with considerable statistical and empirical support.

Beyond this general objection, Scholl and Leslie object that ToMM/SP predicts the meta-analytic findings equally as well as a conceptual change account, and that ToMM/SP's prediction of the findings are actually stronger than alternatives such as theory-theory. But, on closer inspection, these specific objections fail. Two examples follow.

1. Scholl and Leslie claim that their modular, early competence model does not require above-chance performance in young children, and hence is consistent with the meta-analytic pattern of results. So far so good. But, crucially, they go on to argue quite clearly (p. 699) that "the selection processing model predicts . . . that older children will tend to be helped more" by task manipulations that reduce the demand for inhibition. That is, they predict interactions of the type that were found for the variable temporal marker in Figure 8 of our article. But, they predict that type of interaction especially for manipulations that reduce the demand for inhibition. However, the meta-analysis straightforwardly fails to support such a prediction. This type of interaction appeared only once, for temporal marker, a variable that is not arguably related to inhibitory factors of the type that SP posits. Moreover, those variables that are arguably related to inhibitory factors—real presence and salience (as we explained in our discussion of executive function accounts)—did not provide even a hint of this predicted interaction. The failure to demonstrate this type of interaction pattern for other factors is not a matter of ceiling effects masking such predicted findings, as Scholl and Leslie imply. Figure 8 shows that there is plenty of room in the data for such interactions to appear; indeed, the key interaction did appear, for temporal marker. Such an interaction could also then, potentially, appear for each of the many variables considered in our analyses, but it did not.

2. Scholl and Leslie believe that, empirically, theory-of-mind acquisition is universal, has a consistent timetable, and manifests itself in effortless computa-

tion of mental states underlying behavior. They claim that both modular accounts and theory-theory accounts are consistent with such empirical data. However, they claim that ToMM/SP provides a stronger account because it "must predict these features," whereas a theory-theory conceptual change account, they claim, could (with "excessive flexibility") predict these features or equally predict their reverse. Scholl and Leslie are correct that both accounts predict universal acquisition, and in doing so both accounts receive support from the meta-analytic finding that demonstrates a developing understanding of false beliefs in all the countries/communities investigated thus far. However, consider further the issue of timetables. Leslie's modular account does indeed predict a consistent timetable across most individuals (except those in impaired populations, such as individuals with autism). But a theory-theory account does not predict such a rigid maturational timetable. Instead, such an account predicts a consistent trajectory, but a trajectory evident in different developmental timetables, across individuals and cultural communities. Theory-theory predicts that different timetables would result from differences in individuals' experiences across their varied lives and societies, and hence differences in accumulating the type of evidence that will lead to theory change. Again, the meta-analytic findings provide data that speak to Scholl and Leslie's explicit predictions. As shown especially in Figures 7 and 9 of our article, the data were not consistent with predictions of consistent timetables; they were, however, consistent with predictions of consistent trajectories evidenced in quite different timetables.

Thus, in both of these ways, among others, the meta-analytic data are inconsistent with predictions of the ToMM/SP account.

CONCLUSION: MORE REASONS TO FAVOR A CONCEPTUAL CHANGE MODEL OVER THE ToMM/SP MODULAR THEORY

In several places in our discussion of the meta-analytic findings, we pointed out that a focus on false-belief performance alone insufficiently constrains theorizing. Scholl and Leslie agree, at least in part, because in their commentary they cite research comparing individuals with autism versus normally developing children, and cite research comparing understanding of false beliefs versus understanding of false photographs. Continuing in this vein, we end by briefly citing research comparing children's conceptions of desires versus beliefs. This comparison is relevant because the ToMM/SP account clearly states (1) that ToMM processes behaviors and computes the mental states that contributed to them (e.g., mental states such as beliefs and desires), and (2) that SP is a general executive process that applies to reasoning about beliefs but also applies in many other situations, such as reasoning about desires. A comparison between desires and beliefs is also one that Astington (2001) in her commentary advocates as important. We agree.

It is now a well-established fact that young children understand mental states of desire and emotion at an age when they fail to reason comparably about beliefs and false beliefs (Bartsch & Wellman, 1995; Flavell, Flavell, Green, & Moses, 1990; Gopnik & Slaughter, 1991; Wellman & Woolley, 1990). For example, an understanding of desires and emotions—a person's wants and preferences—as subjective and as distinguished from overt behaviors seems in place before 2 years. To illustrate, Repacholi and Gopnik (1997) had 18-month-old toddlers taste two snacks: broccoli and goldfish crackers. Then an adult, facing the child, tasted each snack, saying "Mm" and smiling to one snack, and saying "Eww" and frowning to the other. In a match condition the adult liked the crackers and disliked the broccoli, matching the child's preference. In a mismatch condition she liked the broccoli instead. When the adult then held her hand halfway between the two snacks and said "I want some more, can you give me some more?," 18-month-olds overwhelmingly gave the adult more of what she, the adult, had liked. In doing so in the mismatch condition, these children demonstrated an understanding of desires as subjective, realizing that the adult wanted broccoli, contrary to their own preference for crackers.

Perhaps, as Leslie claims, a theory-of-mind module is active at this early age and thus computes for the child the mental states underlying the adult's behavior—in this case computing the person's desires. If so, however, and ToMM computes both desires and beliefs (as also claimed by Leslie), why does ToMM successfully compute desire at this tender age but, apparently, fails to compute beliefs and false beliefs at the same time? Here is where, within the ToMM/SP account, SP comes in. As with other executive functioning claims that we termed executive performance accounts (and Moses, in his commentary, terms executive function expression accounts),[2] Leslie claims that to correctly pass false-belief tasks, children must inhibit their prepotent tendencies to say or

[2] More generally, we are aware that executive functioning is a crucial feature of SP; all the arguments we advance against executive function performance accounts in that section of our article, apply against the ToMM/SP account as well.

judge one thing, in order to give a different response. In this type of account, developmental increases in executive function capacities would then allow children to increasingly inhibit their incorrect judgments and responses and, thus, as they get older, allow them to correctly reveal their underlying understanding of beliefs as well as desires. However, executive function developments alone seem insufficient to explain the differences in children's judgments of desires versus beliefs.

For example, in a false-belief task, it seems clear that children have to inhibit a prepotent consideration of what they think to be true in order to consider what the other person thinks to be true. Similarly, in Repacholi and Gopnik's (1997) broccoli–crackers task, children have to inhibit their own prepotent preference for the crackers in order to attribute to the adult a preference for broccoli. Although these tasks seem to require comparable inhibitory control, children judge correctly for desires at 18 months, but only do so for beliefs at about 4 years of age.

Perhaps, Leslie could counter, children have only a special difficulty in inhibiting an answer they know to be true (i.e., young children have no general SP problem in inhibiting their own mental states, but only in inhibiting responses that are true). If that were so, it could account for the data discussed thus far. But, if the key difficulty is inhibiting true mental states, then belief tasks without that special difficulty should be correctly solved just as desire tasks are, at an early age. So, now consider a belief task more closely comparable to the Repacholi and Gopnik task—one that requires only judgments that someone else's belief is different from one's own (without requiring the judgment that the other's belief is demonstrably not true). For example, suppose children do not know what is in a box, but think it holds a doll. To judge that someone else thinks the box has a toy truck instead children may need to inhibit their own beliefs (just as in the Repacholi and Gopnik task they must inhibit their own preference for goldfish), but they do not have to inhibit a known true answer (because they don't know the true contents). Moses (2001) mentions this type of belief task in his commentary. We will call this a diverse-belief task, in contrast to a false-belief task. Children do not pass such diverse-belief tasks until about age 3 or 3½ years (Flavell et al., 1990; Wellman & Bartsch, 1989; Wellman, Hollander, & Schult, 1996). Indeed, even when tasks are carefully designed to have closely comparable demands and formats, 2- and 3-year-old children can attribute diverse desires to self versus others, but not diverse beliefs (Flavell et al., 1990; Wellman & Woolley, 1990). Since these tasks are remarkably comparable in SP executive functioning requirements, and since ToMM supposedly can deal with both beliefs and desires, why is there such a substantial lag in correct judgments? This lag remains a mystery in a ToMM/SP account. There is no mystery, however, if we conclude instead that there is a conceptual change in children's understanding of persons and their mental states, a change that progresses from an earlier understanding of desires (but not beliefs) to a later understanding of desires along with beliefs and false beliefs.

In short, although executive functioning skills undoubtedly impact children's understanding of mental states, executive functioning developments fall short of explaining the change apparent in 3-, 4-, and 5-year-olds' understanding of beliefs and false beliefs. (For a more detailed presentation of these and other arguments see Wellman, in press.) This shortfall applies to the ToMM/SP account as well as others, and, we believe, helps to explain why Moses, in his commentary and elsewhere (Carlson & Moses, in press), prefers what he now terms executive function emergence accounts over executive function expression accounts. Executive function emergence accounts accept that an important conceptual change takes place in the years from 3 to 5 or so (and then claim that advances in executive functioning contribute to the emergence of this conceptual change).

To summarize, then, we agree with Scholl and Leslie (and also with Moses) that the meta-analytic findings do not definitively rule out early competence accounts, but they do make extant versions of such accounts increasingly unlikely. In particular, the meta-analytic findings argue against executive function expression accounts, such as ToMM/SP. Moreover, Scholl and Leslie articulate two specific predictions of their account: that task manipulations that attenuate inhibitory demands should differentially advantage older children, and that theory-of-mind developments should occur with consistent timetables. Both of these specific predictions are clearly contradicted, not supported, by the meta-analytic findings. Finally, other findings, for example, those comparing children's reasoning in tasks comparing judgments of desire versus belief, favor a conclusion of conceptual change in the general sense (and favor a more specific theory-theory account as well, we believe). Research comparing children's understanding of beliefs versus desires goes beyond the scope of the studies specifically encompassed in the meta-analysis. But, to reiterate our earlier conclusions, we hope that the meta-analytic findings will diminish research narrowly focused on false-belief performance alone, and will encourage research that focuses on patterns of

performance across a theoretically well-chosen package of comparison tasks.

ADDRESSES AND AFFILIATIONS

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